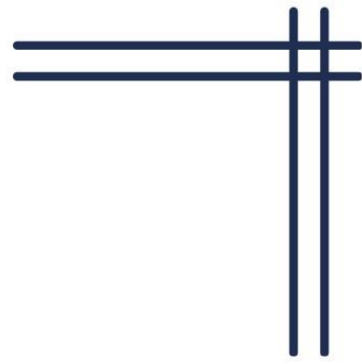




COLLEGE OF
ENGINEERING
POONJAR

(Managed by IHRD, Govt. of Kerala)



GROUP 4

PROJECT

SENTINEL_FUSION

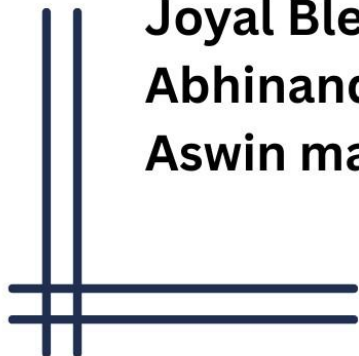
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SENTINEL FUSION : AN AI-POWERED DIGITAL INTELLIGENCE FUSION PLATFORM FOR DIGITAL FORENSICS & INVESTIGATIONS

Project-Based Placement Readiness Programme

Training and Placement Cell

Team Number: IVTeam Name/Repository: Team-4-Sentinel_fusion

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Introduction:

The rapid growth of digital technologies has transformed the way individuals communicate, store information, and conduct daily activities. Smartphones, computers, cloud platforms, social media, messaging applications, and Internet-connected devices generate vast amounts of digital data that can serve as valuable evidence during criminal investigations and cybersecurity incidents. As a result, digital forensics has become an essential discipline for identifying, collecting, preserving, analyzing, and presenting digital evidence in a legally admissible manner.

Modern investigations often involve thousands of digital artifacts, including images, videos, audio recordings, chat messages, documents, browser histories, GPS locations, device metadata, and cloud backups. Analyzing these diverse sources manually is a time-consuming and resource-intensive process. Investigators must review large volumes of evidence, identify relevant information, establish relationships between entities, reconstruct timelines, and document findings while ensuring the integrity and authenticity of digital evidence. The increasing complexity and volume of digital data make traditional forensic workflows insufficient for timely investigations.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), Computer Vision, Natural Language Processing (NLP), and Graph Analytics provide new opportunities to enhance digital forensic investigations. These technologies can automate repetitive tasks such as metadata extraction, optical character recognition (OCR), speech transcription, content classification, and relationship analysis. AI can also assist investigators by identifying hidden connections, reconstructing chronological events, detecting manipulated or AI-generated media, and generating explainable summaries that support informed decision-making.

Sentinel Fusion is proposed as an AI-powered Digital Intelligence Fusion Platform that integrates multiple AI technologies into a unified investigative environment. The platform is designed to consolidate evidence from diverse digital sources, perform intelligent analysis, visualize relationships between entities, reconstruct investigation timelines, and generate explainable investigative insights while maintaining strict human oversight. Rather than replacing investigators or certified forensic tools, Sentinel Fusion functions as an intelligent investigative assistant that accelerates evidence analysis and supports investigators in identifying high-priority leads more efficiently.

The platform emphasizes evidence integrity through secure evidence handling, metadata preservation, chain of custody tracking, role-based access control, and immutable audit logs. By combining explainable AI with modern forensic practices, Sentinel Fusion aims to reduce investigation time, improve analytical accuracy, and

enhance the overall efficiency of digital forensic investigations while ensuring transparency, accountability, and compliance with established forensic principles.

This project contributes to the growing field of AI-assisted digital forensics by providing a scalable, secure, and intelligent platform capable of transforming fragmented digital evidence into actionable investigative intelligence. The proposed system seeks to improve investigator productivity, facilitate faster case progression, and support effective decision-making without compromising the reliability or admissibility of digital evidence.

Background and Motivation: The increasing dependence on digital devices and online platforms has resulted in a significant rise in the amount of digital evidence available during investigations. Information such as images, videos, chat messages, emails, browser history, location data, and device metadata has become essential for understanding criminal activities and cyber incidents. As the volume and complexity of digital evidence continue to grow, investigators face challenges in efficiently analyzing and correlating information from multiple sources.

Existing digital forensic tools primarily focus on evidence acquisition and analysis of individual data sources. However, investigators often need to manually connect related evidence, reconstruct event timelines, identify relationships between entities, and generate comprehensive investigation reports. This manual process is timeconsuming, resource-intensive, and increases the possibility of overlooking critical information.

Recent advancements in Artificial Intelligence, Computer Vision, Natural Language Processing, and Graph Analytics provide new opportunities to improve digital forensic investigations. These technologies can automate repetitive tasks, assist in identifying meaningful patterns, and present investigation findings in a more organized and explainable manner.

The motivation behind Sentinel Fusion is to develop an AI-assisted digital intelligence platform that simplifies evidence analysis by integrating multiple evidence sources into a unified investigation environment. The proposed system aims to support investigators through intelligent evidence correlation, automated timeline reconstruction, explainable AI insights, and secure evidence management while ensuring that all investigative decisions remain under human oversight.

By reducing manual effort and improving investigative efficiency, Sentinel Fusion seeks to accelerate digital forensic investigations without compromising evidence integrity, transparency, or legal reliability.

Problem Statement: The rapid increase in digital devices and online communication has led to an exponential growth in the volume of digital evidence available during investigations. Investigators must analyze diverse data sources such as images, videos, documents, chat messages, browser histories, GPS data, and device metadata to uncover relevant information. Processing this evidence manually is often time-consuming, labor-intensive, and prone to human error.

Although existing digital forensic tools provide capabilities for evidence acquisition and analysis, they generally operate on individual data sources and offer limited support for automatically correlating evidence, reconstructing timelines, identifying hidden relationships, and detecting manipulated or AI-generated media. As a result, investigators spend significant time performing repetitive tasks, which can delay investigations and increase the risk of missing critical evidence.

There is a need for an intelligent platform that can consolidate heterogeneous digital evidence, automate routine analytical tasks, and provide explainable insights while preserving evidence integrity and ensuring that all investigative decisions remain under human oversight. Addressing these challenges can significantly improve the efficiency, accuracy, and effectiveness of modern digital forensic investigations.

Objectives: The primary objective of Sentinel Fusion is to develop an AI-powered Digital Intelligence Fusion Platform that assists digital forensic investigators by integrating, analyzing, and visualizing digital evidence from multiple sources in a secure and explainable manner.

The specific objectives of the project are:

- To develop a unified platform for uploading and managing multiple types of digital evidence, including images, videos, audio, documents, and chat exports.
- To automate metadata extraction and content analysis using Artificial Intelligence techniques such as OCR, Computer Vision, and Speech-to-Text.
- To correlate evidence from multiple sources and identify relationships between entities using graph-based analysis.
- To reconstruct investigation timelines by organizing events based on extracted timestamps and metadata.
- To detect manipulated or AI-generated media and provide authenticity assessment to support evidence verification.
- To generate explainable AI-assisted investigation summaries and reports that support decision-making while maintaining human oversight.
- To ensure evidence integrity through secure storage, hash verification, role-based access control, and audit logging.

Literature Review: Digital forensics has become an essential field in modern criminal investigations due to the increasing reliance on digital devices and online communication. Various forensic tools such as Autopsy, FTK (Forensic Toolkit), and Magnet AXIOM are widely used to acquire, preserve, and analyze digital evidence from computers, mobile devices, and storage media. These tools provide reliable capabilities for evidence extraction, file recovery, metadata analysis, and reporting. However, they primarily focus on individual evidence analysis and require significant manual effort to correlate information across multiple data sources.

Recent research has explored the integration of Artificial Intelligence (AI) and Machine Learning (ML) into digital forensic investigations. AI techniques such as Computer Vision, Natural Language Processing (NLP), and Speech Recognition have demonstrated the ability to automate tasks including image classification, optical character recognition (OCR), audio transcription, and document analysis. These technologies reduce manual workload and enable investigators to process large volumes of digital evidence more efficiently.

Graph-based intelligence systems have also gained attention for representing relationships among people, devices, locations, and digital artifacts. By modeling evidence as interconnected entities, graph databases help investigators discover hidden connections and reconstruct investigative timelines. Additionally, advances in Large Language Models (LLMs) have enabled automatic summarization of forensic findings and generation of explainable reports, improving the efficiency of documentation and evidence review.

Despite these advancements, most existing solutions focus on specific aspects of digital forensics rather than providing a unified platform that integrates evidence ingestion, AI-assisted analysis, relationship mapping, timeline reconstruction, synthetic media detection, and explainable reporting. Furthermore, many systems provide limited transparency in AI-generated results, making it difficult for investigators to validate findings.

Sentinel Fusion addresses these limitations by integrating multiple AI technologies into a single Digital Intelligence Fusion Platform. The proposed system combines automated evidence analysis, metadata extraction, graphbased relationship visualization, timeline reconstruction, deepfake detection, and explainable AI-assisted reporting while ensuring evidence integrity and maintaining human oversight throughout the investigation process.

Innovation: The key innovation of Sentinel Fusion lies in its ability to integrate multiple AI technologies into a single Digital Intelligence Fusion Platform that supports digital forensic investigations. Unlike conventional forensic tools that primarily focus on evidence acquisition and isolated analysis, Sentinel Fusion combines evidence correlation, timeline reconstruction, graph-based relationship visualization, AI-assisted content analysis, and synthetic media detection within a unified investigation environment.

The platform leverages Artificial Intelligence techniques, including Computer Vision, Natural Language Processing (NLP), Optical Character Recognition (OCR), Speech-to-Text, and Graph Analytics, to automate repetitive investigative tasks and transform fragmented digital evidence into connected and meaningful intelligence. This integrated approach enables investigators to identify hidden relationships, prioritize investigative leads, and gain a comprehensive understanding of cases more efficiently.

A distinguishing feature of Sentinel Fusion is its emphasis on Explainable AI (XAI). Every AI-generated insight is accompanied by supporting evidence and confidence scores, enabling investigators to understand and verify the reasoning behind each recommendation. Furthermore, the platform follows a human-in-the-loop approach, ensuring that AI serves as an investigative assistant while all critical decisions remain under the control of authorized investigators.

By combining intelligent automation, explainable analytics, secure evidence management, and human oversight, Sentinel Fusion offers an innovative and scalable solution for modern digital forensic investigations.

Proposed Methodology: The proposed system, Sentinel Fusion, follows a structured workflow to transform fragmented digital evidence into actionable investigative intelligence. The platform integrates Artificial Intelligence, metadata analysis, graph analytics, and secure evidence management to assist investigators throughout the investigation process while ensuring evidence integrity and human oversight.

The methodology consists of the following stages:

1. Evidence Collection and Ingestion

Digital evidence, including images, videos, audio recordings, documents, chat exports, and forensic data, is securely uploaded to the platform. During ingestion, each file is assigned a unique identifier, and a SHA-256 hash is generated to ensure evidence integrity. Chain of custody information is also recorded for audit purposes.

2. Evidence Processing

The uploaded evidence undergoes automated preprocessing using specialized AI components. Metadata is extracted from files, OCR is applied to images and documents, speech-to-text converts audio into text, and computer vision models analyze multimedia content to identify relevant objects, scenes, and textual information.

3. Intelligence Fusion

Information extracted from different evidence sources is correlated to identify relationships among people, devices, locations, accounts, and events. A graphbased intelligence engine constructs an interactive relationship network, enabling investigators to discover hidden connections across multiple evidence sources.

4. Timeline Reconstruction

Extracted timestamps, metadata, communication records, and location information are organized chronologically to reconstruct the sequence of events. This provides investigators with a clear visualization of the investigation timeline.

5. AI-Assisted Analysis

The processed evidence is analyzed using Artificial Intelligence to generate investigation summaries, identify high-risk content, detect manipulated or AI-generated media, and prioritize investigative leads. All AI-generated findings are accompanied by confidence scores and supporting evidence to ensure transparency and explainability.

6. Investigator Dashboard and Report Generation

The investigation results are presented through an interactive dashboard featuring evidence summaries, relationship graphs, timelines, and risk indicators. Investigators can review AI-generated insights, verify findings, and generate comprehensive investigation reports in standard formats while maintaining full control over investigative decisions.

Technology Stack: The proposed system is developed using a modern, AI-oriented technology stack that ensures scalability, security, and efficient processing of digital evidence. The selected technologies support AI integration, secure evidence management, and interactive data visualization required for digital forensic investigations.

Component	Technology	Purpose
Frontend	Next.js (React)	Develops an interactive and responsive user interface for investigators.
Styling	Tailwind shadcn/ui	CSS, Creates a modern and consistent dashboard interface.
Backend	FastAPI (Python)	Handles API services, business logic, and AI integration.
Authentication	Clerk	Provides secure user authentication, MFA, and role-based access control.
Database	PostgreSQL	Stores user information, case details, evidence metadata, and audit logs.
Graph Database	Neo4j	Represents relationships among entities for intelligence analysis.
Vector Database	Qdrant	Enables semantic search and AI-powered evidence retrieval.
Cache & Queue	Redis	Supports caching and background processing of AI tasks.
Object Storage	AWS S3 / MinIO	Securely stores multimedia evidence files.
OCR Engine	PaddleOCR	Extracts text from images and scanned documents.
Metadata Extraction Component	ExifTool Technology	Retrieves metadata from digital evidence files.
Speech Recognition	OpenAI Whisper	Converts audio recordings into searchable text.

Vision AI	Florence-2	/ Analyzes images and videos for object and scene understanding.
	Gemma Vision	
AI Workflow	LangGraph	Coordinates AI processing pipelines and investigation workflows.
Large Language Model	GPT-5.5 API	Generates explainable investigation summaries and AI-assisted insights.
Search Engine	OpenSearch	Provides fast keyword and metadatabased evidence search.
Deployment	Docker, Kubernetes	Enables containerized deployment and scalable infrastructure.
Monitoring	Grafana, Prometheus, Sentry	
		Monitors system performance and reliability.

Technology Selection

The chosen technology stack combines modern web development frameworks, secure backend services, AI processing tools, and scalable databases to support efficient digital forensic investigations. These technologies enable Sentinel Fusion to securely manage digital evidence, automate analytical tasks, visualize relationships between entities, and generate explainable investigative insights while maintaining system scalability and performance.

Implementation Plan / Modules: The implementation of Sentinel Fusion is divided into several functional modules, each responsible for a specific aspect of the digital forensic investigation process. This modular architecture ensures scalability, maintainability, and efficient integration of AI-powered services.

Module 1: User Authentication and Case Management

This module provides secure user authentication and authorization using rolebased access control (RBAC). It enables investigators to create, manage, and organize investigation cases while maintaining secure access to case-related information.

Module 2: Evidence Ingestion and Management

This module allows investigators to upload and securely store multiple types of digital evidence, including images, videos, audio files, documents, and chat exports. It also performs file integrity verification through SHA-256 hashing and maintains chain of custody records.

Module 3: AI-Based Evidence Analysis

The uploaded evidence is processed using Artificial Intelligence techniques such as Optical Character Recognition (OCR), Computer Vision, Speech-to-Text, and metadata extraction. The extracted information is used to identify relevant entities and generate meaningful insights.

Module 4: Intelligence Fusion and Timeline Reconstruction

This module correlates information extracted from different evidence sources to identify relationships among people, devices, locations, and events. It also reconstructs a chronological timeline of activities to help investigators understand the sequence of events.

Module 5: Investigation Dashboard and Reporting

This module presents investigation results through an interactive dashboard featuring evidence summaries, relationship graphs, timelines, AI-generated findings, and risk indicators. It also supports the generation of comprehensive investigation reports for documentation and review.

Implementation Timeline: The development of Sentinel Fusion is planned in a phased manner to ensure systematic implementation, testing, and evaluation. Each phase focuses on specific milestones, enabling the team to monitor progress and complete the project within the proposed schedule.

Phase	Duration	Key Activities	Deliverables
Phase 1: Planning & Setup	Week 1	Finalize requirements, define architecture, set up GitHub, create database schema, configure development environment, assign team roles	PRD, Tech Stack, Git Repository, Project Skeleton
Phase 2: Core Development	Week 2	Develop frontend dashboard, backend APIs, authentication, database models, evidence upload module	Authentication, Dashboard, Evidence Management APIs
Phase 3: AI & Intelligence Engine	Week 3	Implement OCR, metadata extraction, speech transcription, image analysis, relationship mapping, deepfake detection prototype	AI Processing Pipeline, Intelligence Fusion Engine
Phase 4: Integration & Analytics	Week 4	Integrate frontend with backend and AI services, build timeline reconstruction, risk assessment, relationship graph, investigation dashboard	Fully Integrated Application
Phase 5: Testing & Optimization	Week 5	Functional testing, performance optimization, security validation, bug fixing, UI refinement, report generation	Stable MVP, Performance Improvements
Phase 6: Deployment & Documentation	Week 6	Docker deployment, CI/CD setup, monitoring, documentation, demo video, presentation preparation	Production-ready MVP, Documentation, Final Presentation

Expected Outcomes: The successful implementation of Sentinel Fusion is expected to provide an AI-powered digital intelligence platform that enhances the efficiency and effectiveness of digital forensic investigations. The anticipated outcomes of the project include:

- Development of a secure and unified platform for managing and analyzing digital evidence from multiple sources.
- Automated extraction of metadata and analysis of images, videos, audio, documents, and chat data using AI techniques.
- Intelligent correlation of evidence to identify relationships between people, devices, locations, and digital artifacts.
- Automatic reconstruction of investigation timelines to help investigators understand the sequence of events.
- Detection of manipulated or AI-generated media to assist in evidence authenticity verification.
- Generation of explainable AI-assisted investigation summaries and comprehensive reports.
- Improved investigator productivity by reducing manual evidence analysis and accelerating case progression.
- Maintenance of evidence integrity through secure storage, hash verification, role-based access control, and audit logging.

The proposed system is expected to serve as an intelligent decision-support platform that assists investigators in analyzing complex digital evidence while ensuring transparency, security, and human oversight throughout the investigation process.

Future Scope: Sentinel Fusion has significant potential for future enhancements to address the evolving needs of digital forensic investigations. As digital evidence continues to increase in volume and complexity, the platform can be expanded with advanced AI capabilities and collaborative investigation features.

The future scope of the project includes:

- Integration with certified digital forensic tools for seamless evidence import and analysis.
- Multi-agency collaboration with secure case and evidence sharing.
- Advanced Natural Language Search for querying evidence using conversational language.
- Enhanced AI models for improved deepfake, voice cloning, and synthetic media detection.
- Real-time monitoring and analysis of digital evidence from authorized sources.
- Geospatial visualization of evidence and GPS data using interactive maps.
- Mobile application for secure access to investigation dashboards and case updates.
- Predictive analytics to identify investigation patterns and assist in lead prioritization.
- Cloud-native deployment for improved scalability, high availability, and distributed processing.

With continuous advancements in Artificial Intelligence and digital forensic technologies, Sentinel Fusion can evolve into a comprehensive Digital Intelligence Fusion Platform that supports faster, more accurate, and explainable investigations while maintaining evidence integrity, security, and human oversight.

References: [1] E. Casey, *Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet*, 3rd ed. Burlington, MA, USA: Academic Press, 2011.

- [2] B. Carrier, *File System Forensic Analysis*. Boston, MA, USA: Addison-Wesley Professional, 2005.
- [3] National Institute of Standards and Technology (NIST), *Guide to Integrating Forensic Techniques into Incident Response*, NIST Special Publication 800-86, Gaithersburg, MD, USA, 2006.
- [4] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [5] H. Garcia-Molina, J. D. Ullman, and J. Widom, *Database Systems: The Complete Book*, 2nd ed. Pearson Education, 2008.
- [6] OpenAI, "GPT API Documentation." Available: <https://platform.openai.com/docs>
- [7] Neo4j, "Neo4j Graph Database Documentation." Available: <https://neo4j.com/docs/>
- [8] FastAPI, "FastAPI Documentation." Available: <https://fastapi.tiangolo.com/>
- [9] Next.js, "Next.js Documentation." Available: <https://nextjs.org/docs>
- [10] PaddleOCR, "PaddleOCR Documentation." Available: <https://github.com/PaddlePaddle/PaddleOCR>
- [11] ExifTool, "ExifTool by Phil Harvey." Available: <https://exiftool.org/>
- [12] OpenAI, "Whisper Speech Recognition." Available: <https://github.com/openai/whisper>